“Mental Health Analysis Using Machine Learning Models”

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(Specialization in Artificial Intelligence & Machine Learning / Data Science)

Submitted by

M. V. S. Satya Mrudula (2023000597)
G. Rishikar Reddy (2023001672)
D. Vivek (2023002218)
T. S. L. Priyanka (2023004539)

Under the esteemed guidance of
Dr. M. Ramakrishna Murty, Professor

Department of Artificial Intelligence and Data Science
GITAM School of CSE, GITAM (Deemed to be University),
Visakhapatnam
2026

DECLARATION

I hereby declare that the project report entitled “Mental Health Analysis Using Machine Learning Models” is an original work done in the Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University), submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering (AI&ML/DS). The work has not been submitted to any other college or University for the award of any degree or diploma.

Date:

Registration No.	**Name**	**Signature**
2023000597	M. V. S. Satya Mrudula	
2023001672	G. Rishikar Reddy	
2023002218	D. Vivek	
2023004539	T. S. L. Priyanka	

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING/ DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

GITAM School of CSE

GITAM (Deemed to be University)



CERTIFICATE

This is to certify that the project report entitled "Mental Health Analysis Using Machine Learning Models" is a bonafide record of work carried out by M. V. S. Satya Mrudula (2023000597), G. Rishikar Reddy (2023001672), D. Vivek (2023002218) and T. S. L. Priyanka (2023004539) students submitted in partial fulfillment of requirement for the award of degree of Bachelor of Technology in Computer Science and Engineering (AI&ML/DS).

Date :

Project Guide Head of the Department

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, Dr. M. Ramakrishna Murty, Professor, Department of Artificial Intelligence and Data Science, GITAM School of CSE, for his valuable guidance, constructive feedback and continuous encouragement throughout this capstone project. His suggestions during the project reviews, particularly on broadening the data sources and selecting up-to-date models, have shaped the direction of this work.

We thank the Head of the Department and the faculty members of the Department of Artificial Intelligence and Data Science for providing the academic environment and resources needed to carry out this project. We also thank the project coordinators for organising the review process.

Finally, we are grateful to our families and friends for their patience and support during the project.

TABLE OF CONTENTS

S.No.	Description	Page No.
1.	Abstract	1
2.	Introduction	1
3.	Literature Review	2
4.	Problem Identification & Objectives	3
5.	Existing System	5
6.	Proposed System	5
7.	System Architecture	8
8.	Tools/Technologies Used	13
9.	Conclusion	14
10.	References	15

1. ABSTRACT

Mental health concerns often remain unrecognised until individuals report symptoms or receive a professional evaluation. Early screening can help identify people who may be at risk of depression, anxiety and stress, yet conventional screening relies largely on questionnaires and clinical judgement. This project investigates whether machine-learning models can combine demographic, behavioural, lifestyle, academic or workplace, social and psychological factors to support early screening. The proposed framework processes two data streams separately: structured questionnaire responses are analysed using an FT-Transformer, and psychological text is analysed using DeBERTa. The outputs of the two models are then combined to generate predictions for depression, anxiety and stress, with the aim of explaining contributing factors to support screening and professional decision-making. At the time of this review, both datasets have been collected and the models have been selected. Preprocessing, model development, integration and evaluation are the planned next steps.

2. INTRODUCTION

2.1 Background

Mental health is an essential part of overall well-being, yet depression, anxiety and stress often go unrecognised. Many individuals do not seek help until symptoms become severe, and routine screening is not always available in schools, colleges and workplaces. Early identification of people at risk can therefore support timely referral to professional care.

Machine learning offers a way to support such screening by learning patterns from data that describe a person's circumstances and experiences. Questionnaire responses capture demographic, behavioural, lifestyle, academic or workplace and social factors in a structured form, while psychological text, such as a person's written description of their feelings or thoughts, carries information that a fixed questionnaire may not capture. Analysing these two sources together has the potential to give a more complete picture than either alone.

2.2 Need for the System

Screening that depends on a single questionnaire can miss useful information, and clinical records are not always available outside health services. Faculty, counsellors and workplace support teams need a way to make the screening process more systematic. A system that uses both structured responses and written psychological text, and that gives clear, readable outputs, can support these decisions while leaving the final judgement to trained professionals.

2.3 Justification of the Title

The title “Mental Health Analysis Using Machine Learning Models” describes the main parts of the work. The machine-learning models learn the relationship between the input data and the screening outcomes, rather than relying on fixed scoring rules. The analysis is directed at mental health, specifically the three conditions of depression, anxiety and stress. The word “analysis” reflects that the system examines the inputs to estimate risk, and that its outputs are intended as decision-support information rather than a diagnosis.

2.4 Scope of the Project

The project covers the selection and preparation of the questionnaire and psychological text datasets, the choice of FT-Transformer and DeBERTa as the two analysis models, the combination of their outputs, and the evaluation of the resulting predictions. It does not cover clinical diagnosis, treatment recommendations or the deployment of the system in a live clinical setting. Predictions are intended only to support screening and professional decision-making.

3. LITERATURE REVIEW

3.1 Machine Learning for Depression and Anxiety Prediction

ShamsEldin et al. used psychometric data to predict depression, anxiety and stress with artificial intelligence, showing that these three conditions can be modelled from questionnaire-style inputs [1]. Nemesure et al. developed a machine-learning approach for predicting depression and anxiety from electronic health records, demonstrating that routinely collected clinical data can support risk prediction [2]. Both studies show that prediction is feasible, but each relies on one type of input.

3.2 Prediction Over Time and Explainable Screening

Bailey and Strunk studied the prediction of anxiety and depression over twelve months of the COVID-19 pandemic, indicating that models can be used to follow changes in mental health over time [3]. Ahmed and Ahmed proposed a fast and minimal system to identify depression from smartphone data, emphasising explainability and a lightweight design that can be deployed easily [4]. These works show the value of explainable and practical screening tools.

3.3 Transformer Models for Text and Tabular Data

DeBERTa introduced disentangled attention to improve transformer-based language models, making it a strong choice for understanding the meaning of text [5]. The study “Revisiting Deep Learning Models for Tabular Data” examined deep architectures for structured data and proposed the FT-

Transformer, a transformer-based model suited to questionnaire-style tables [6]. Together, these works motivate the use of modern transformer models for both data types rather than older linear or margin-based classifiers.

3.4 Summary and Research Gap

Table 3.1 summarises the reviewed work and the gap that this project addresses.

Table 3.1 : Summary of reviewed work and the gap addressed

Ref.	Focus	Contribution	Gap addressed in this project
[1]	Psychometric data	AI prediction of depression, anxiety and stress	Single questionnaire-style source; text not used
[2]	Electronic health records	Machine-learning prediction of depression and anxiety	Clinical records are not the input for this project
[3]	Pandemic-period data	Prediction over twelve months	Changes over time are not the focus here
[4]	Smartphone data	Fast, explainable depression detection	Anxiety and stress are not addressed together
[5]	Text models	Disentangled attention for language understanding	Applied here to psychological text for screening
[6]	Tabular models	FT-Transformer for structured data	Applied here to questionnaire responses for screening

The literature shows that prediction of these conditions is established, but the two kinds of input are usually studied separately. This project therefore combines structured questionnaire responses and psychological text in one framework and predicts all three conditions.

4. PROBLEM IDENTIFICATION & OBJECTIVES

4.1 Problem Identification

Mental-health concerns may remain unrecognised until individuals report symptoms or receive professional evaluation. The problem addressed in this project is whether machine-learning models can combine demographic, behavioural, lifestyle, academic or workplace, social and psychological factors to identify individuals who may be at risk of depression, anxiety or stress, and whether they can do so in a way that explains the contributing factors.

4.2 Aim of the Project

The aim is to investigate whether machine-learning models can support early screening for possible depression, anxiety and stress by using relevant demographic, behavioural, lifestyle, academic or workplace, social and psychological factors.

4.3 Objectives

- Analyse questionnaire data using the FT-Transformer model.
- Analyse psychological text data using the DeBERTa model.
- Combine the outputs of both models to generate predictions for depression, anxiety and stress.

4.4 Feedback from Review 1 and Actions Taken

The feedback received from the guide and reviewers in Review 1 [7] was used to shape the current scope. Table 4.1 lists the feedback and the corresponding actions.

Table 4.1 : Review 1 feedback and actions taken

S.No.	Feedback received in Review 1	Action taken
1	Expand the project beyond questionnaire-based analysis.	Added psychological text data as an additional source alongside questionnaire data.
2	Avoid relying on older models such as SVM and Logistic Regression; explore updated models from research papers.	Selected DeBERTa for text analysis and FT-Transformer for questionnaire analysis.
3	Include only information that the team can clearly explain and justify.	Planned two separate models whose outputs are combined to predict depression, anxiety and stress.
4	Progress on data collection.	Collection of both datasets completed.

4.5 Requirement Analysis

The requirements derived from the objectives and from the Review 1 feedback are listed in Table 4.2.

Table 4.2 : Functional requirements of the system

S.No.	Requirement	Purpose
1	Use the benchmark questionnaire and psychological text datasets.	Provide the inputs for both models
2	Inspect data formats, missing values, labels and data quality.	Ensure the data are fit for modelling

S.No.	Requirement	Purpose
3	Preprocess structured questionnaire responses and psychological text.	Prepare each stream for its model
4	Analyse questionnaire data using FT-Transformer.	Produce the questionnaire-based output
5	Analyse psychological text using DeBERTa.	Produce the text-based output
6	Combine model outputs to predict depression, anxiety and stress.	Give one result per condition
7	Provide clear and readable outputs.	Make results usable by screening staff

5. EXISTING SYSTEM

5.1 Current Approaches

Screening for depression, anxiety and stress is commonly carried out with standardised questionnaires that are scored manually or with simple statistical rules. Where machine learning has been applied, many studies use a single data source, typically questionnaire or clinical records, together with conventional classifiers such as Support Vector Machines and Logistic Regression. Table 5.1 summarises these approaches and their drawbacks.

Table 5.1 : Existing approaches and their drawbacks

Existing approach	Main drawback
Standardised questionnaire scoring	Relies on one structured source; written expression of feelings is not used.
Questionnaire-based machine learning with SVM or Logistic Regression	Older models that may not capture complex relationships as effectively as recent architectures.
Clinical-record prediction	Depends on records that are not available for many screening settings.
Single-source prediction	Predictions are not combined with a second, complementary source of information.

In summary, existing approaches usually rely on one data source and an older model, and they rarely combine structured responses with psychological text in a single, readable output.

6. PROPOSED SYSTEM

6.1 Overview

The proposed system extends questionnaire-based analysis by adding psychological text as a second data source. Two separate models are used, each suited to its data type. An FT-Transformer analyses

the structured questionnaire responses, and DeBERTa analyses the psychological text. The outputs of the two models are then combined to generate predictions for depression, anxiety and stress.

6.2 Modules of the Proposed System

- Data validation: checks the questionnaire and text datasets for data formats, missing values, labels and data quality.
- Questionnaire pipeline: preprocesses the structured responses and analyses them with the FT-Transformer.
- Text pipeline: preprocesses the psychological text and analyses it with DeBERTa.
- Integration module: combines the outputs of both models to produce predictions for depression, anxiety and stress.
- Output module: presents the predictions in a clear and readable form.

6.3 Datasets

Two datasets are used in the project, as listed in Table 6.1. Dataset collection is complete. Confirming the labels, checking the data formats and resolving any data-quality issues are part of the next stage.

Table 6.1 : Datasets used in the project

Dataset	Content	Use in the project
Benchmark questionnaire dataset	Responses covering demographic, behavioural, lifestyle, academic or workplace, social and psychological factors	Input to the FT-Transformer for questionnaire analysis
Psychological text dataset	Written psychological text	Input to DeBERTa for text analysis

6.4 Proposed Methodology

The questionnaire and psychological text data are processed through separate pipelines. The FT-Transformer analyses the questionnaire data, DeBERTa analyses the psychological text, and the outputs of the two models are combined to generate predictions for depression, anxiety and stress. Figure 6.1 shows the proposed methodology.

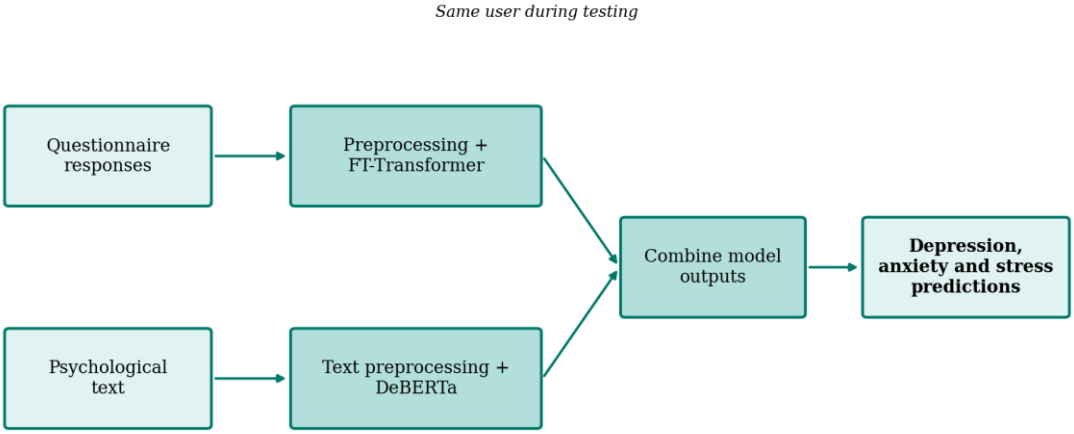


Figure 6.1 : Proposed methodology

6.5 Combination of Model Outputs

During testing, questionnaire responses and psychological text are provided by the same user. Each model produces its own output, and the integration stage combines these outputs into one prediction for each condition. The exact combination method is planned to be chosen after baseline experiments, and it will be reported together with its results.

6.6 Evaluation Plan

The models will be compared using metrics that suit a screening task (Table 6.2). The evaluation will also check for missing values, class imbalance, data leakage and differences in error rates across subgroups.

Table 6.2 : Planned evaluation of the models

Task	Model	Planned metrics
Questionnaire-based prediction of depression, anxiety and stress	FT-Transformer	Accuracy, precision, recall, F1-score, ROC-AUC
Text-based prediction of depression, anxiety and stress	DeBERTa	Accuracy, precision, recall, F1-score, ROC-AUC
Combined prediction	Integration of both model outputs	Same metrics, compared with each single-model output

6.7 Improvement over the Existing System

Table 6.3 compares the existing and the proposed system.

Table 6.3 : Comparison of the existing and the proposed system

Aspect	Existing system	Proposed system
Data used	Single structured source, or clinical records	Questionnaire responses and psychological text
Models	Support Vector Machine, Logistic Regression	FT-Transformer and DeBERTa
Predictions	Usually one condition at a time	Depression, anxiety and stress
Output	Score or category with little context	Clear and readable outputs for screening staff
Decision making	Depends on the individual assessor	Decision support, with professional review of results

7. SYSTEM ARCHITECTURE

7.1 Project Workflow

The implementation follows the workflow shown in Figure 7.1. The questionnaire and psychological text pipelines run separately, and their outputs meet in the combined output stage. Table 7.1 shows the status of each stage at the time of this review.

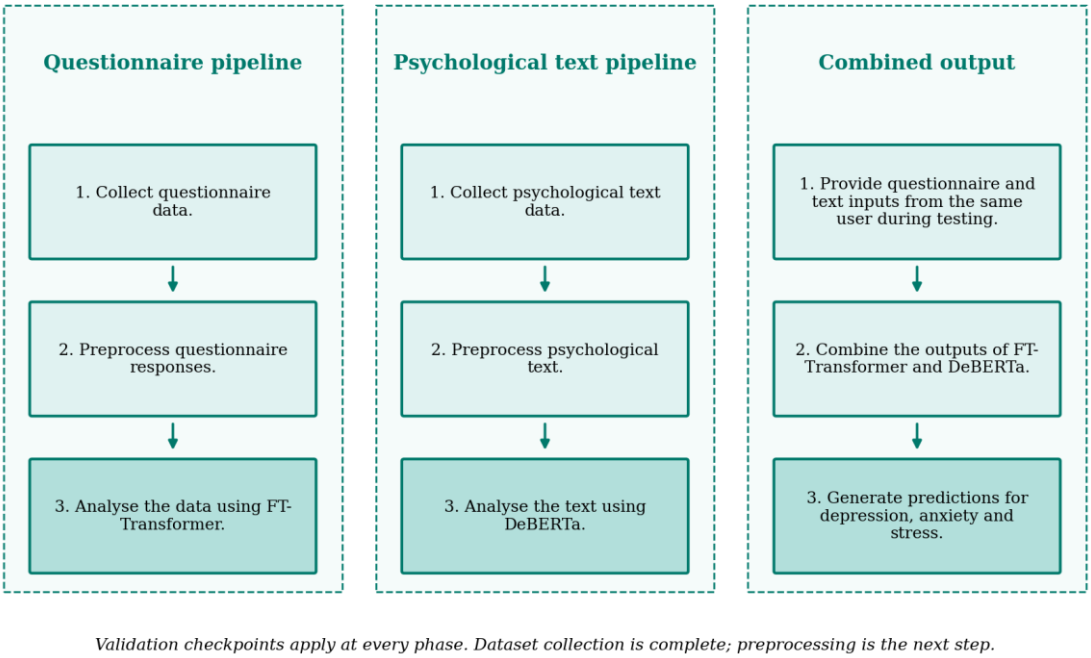


Figure 7.1 : Proposed project workflow

- Validation checkpoints: check the data formats, missing values, labels and data quality before each analysis stage.
- Expected deliverables: prepared questionnaire and text datasets, trained FT-Transformer and DeBERTa models, combined predictions for depression, anxiety and stress, and reproducible results for the next review.

Table 7.1 : Project workflow and status

Stage	Activity	Status
1	Collection of the questionnaire and psychological text datasets	Completed
2	Data inspection and preprocessing of both datasets	Next step
3	Model development with FT-Transformer and DeBERTa	Planned
4	Integration of model outputs	Planned
5	Testing and evaluation	Planned
6	Presentation of results and review	Planned

7.2 Data Flow Diagram

Figure 7.2 shows the conceptual data flow. During testing, the same user provides questionnaire responses and psychological text. The questionnaire responses are preprocessed and passed to the FT-Transformer, and the psychological text is preprocessed and passed to DeBERTa. The two model outputs are then combined to predict depression, anxiety and stress. Table 7.2 describes the components of the data flow.

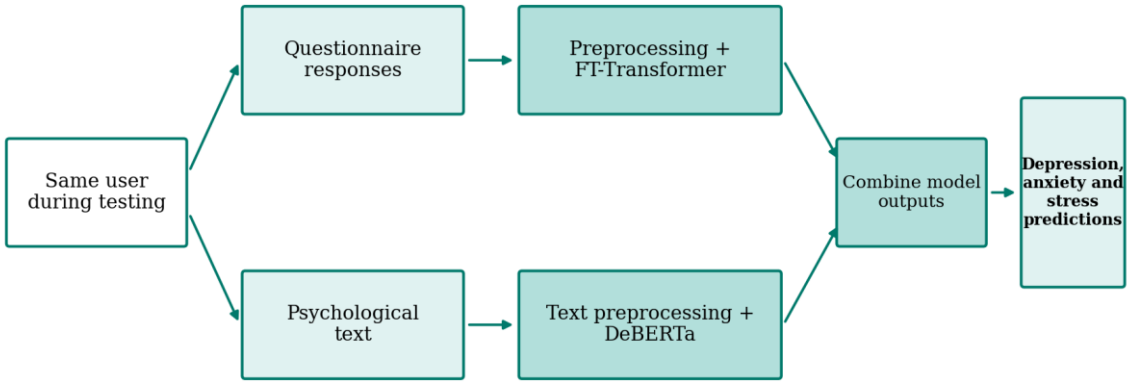


Figure 7.2 : Conceptual data flow diagram of the proposed system

Table 7.2 : Main components of the data flow

Component	Description
Input	Questionnaire responses and psychological text from the same user
Processing	Preprocessing of each data stream and data-quality checks
Prediction	Analysis with the FT-Transformer and DeBERTa
Integration	Combination of the two model outputs
Output	Predictions for depression, anxiety and stress

7.3 Use Case Diagram

Figure 7.3 shows the conceptual use case design. The user is the primary actor. Requesting a mental-health screening includes the analysis of the questionnaire responses with the FT-Transformer, the analysis of the psychological text with DeBERTa, and the combination of both predictions. Table 7.3 lists the use cases.

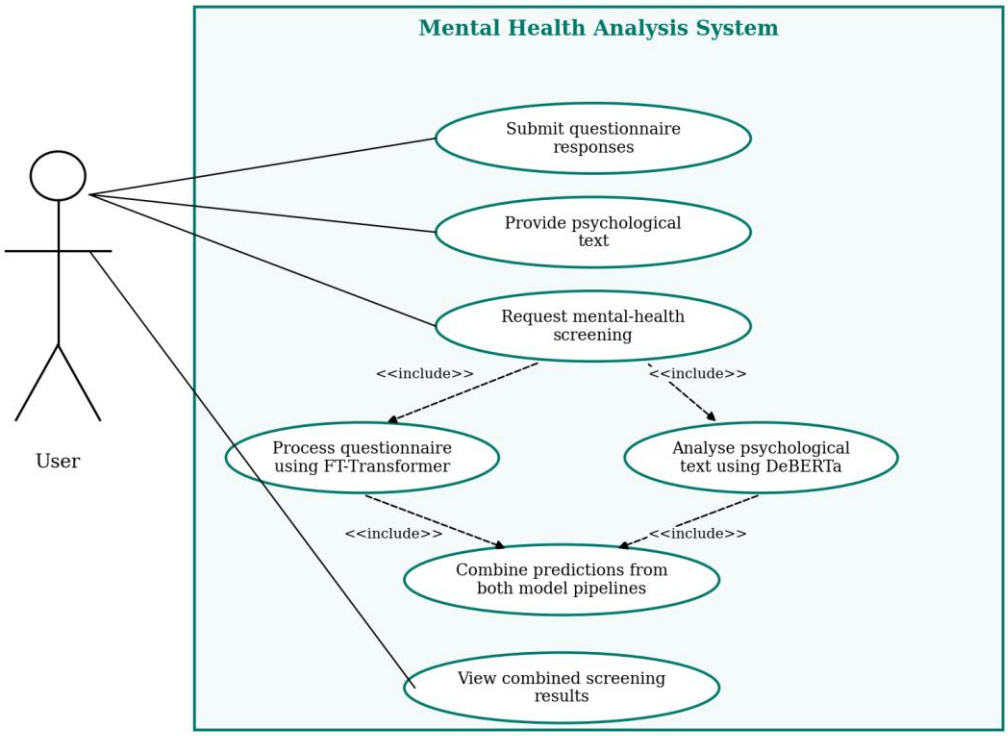


Figure 7.3 : Use case diagram of the proposed system

Table 7.3 : Use cases of the system

Use case	Description
Submit questionnaire responses	The user provides the structured questionnaire responses
Provide psychological text	The user provides written psychological text
Request mental-health screening	The user starts the screening, which includes the two analyses below
Process questionnaire using FT-Transformer	Included in the screening request
Analyse psychological text using DeBERTa	Included in the screening request
Combine predictions from both model pipelines	Included in the screening request
View combined screening results	The user sees predictions for depression, anxiety and stress

7.4 Sequence Diagram

Figure 7.4 shows the interaction between the user, the frontend interface, the two analysis models and the integration module. The user submits the questionnaire responses and the psychological text, and

then requests a screening. The interface sends each input to its model. The two predictions are passed to the integration module, which combines them, and the combined results are returned to the user.

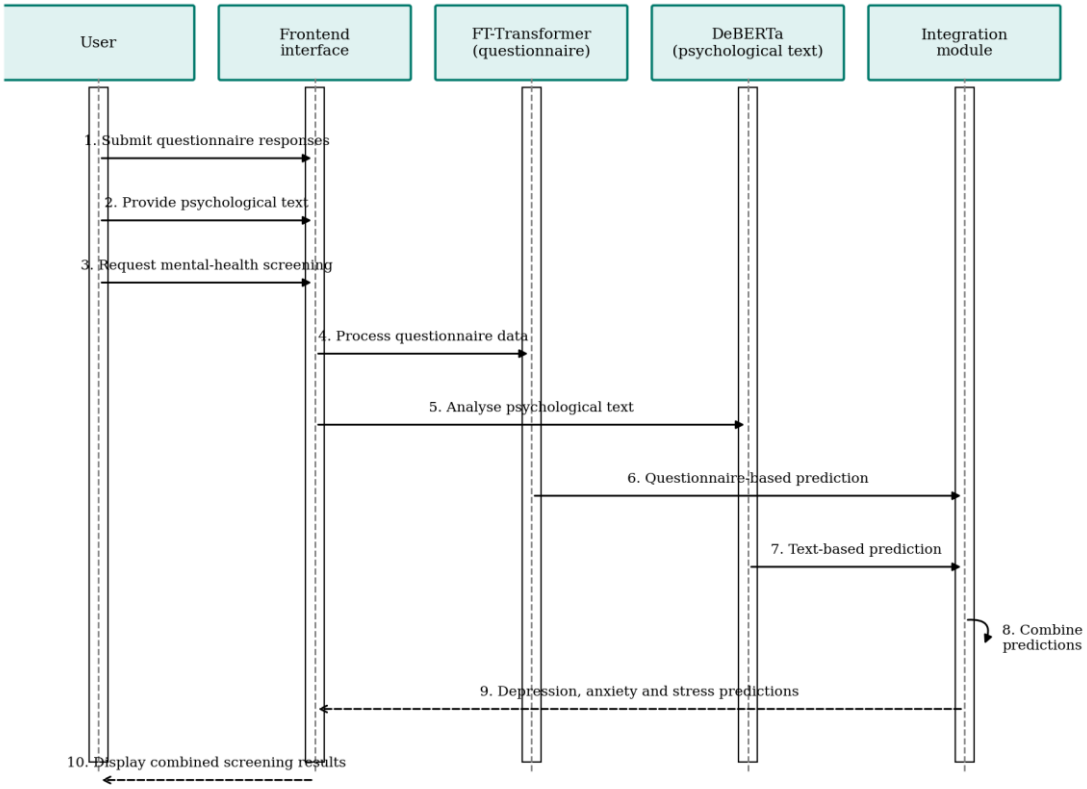


Figure 7.4 : Sequence diagram of the mental-health screening process

7.5 Activity Diagram

Figure 7.5 follows the activity of the screening process across four swimlanes. After the user submits both inputs and requests a screening, the questionnaire responses and the psychological text are preprocessed in parallel. The questionnaire data are checked for validity; if the check fails, the record is flagged for correction. Otherwise, the FT-Transformer analyses the data. DeBERTa analyses the psychological text. The integration stage then combines the two outputs, generates predictions for depression, anxiety and stress, and displays them to the user.

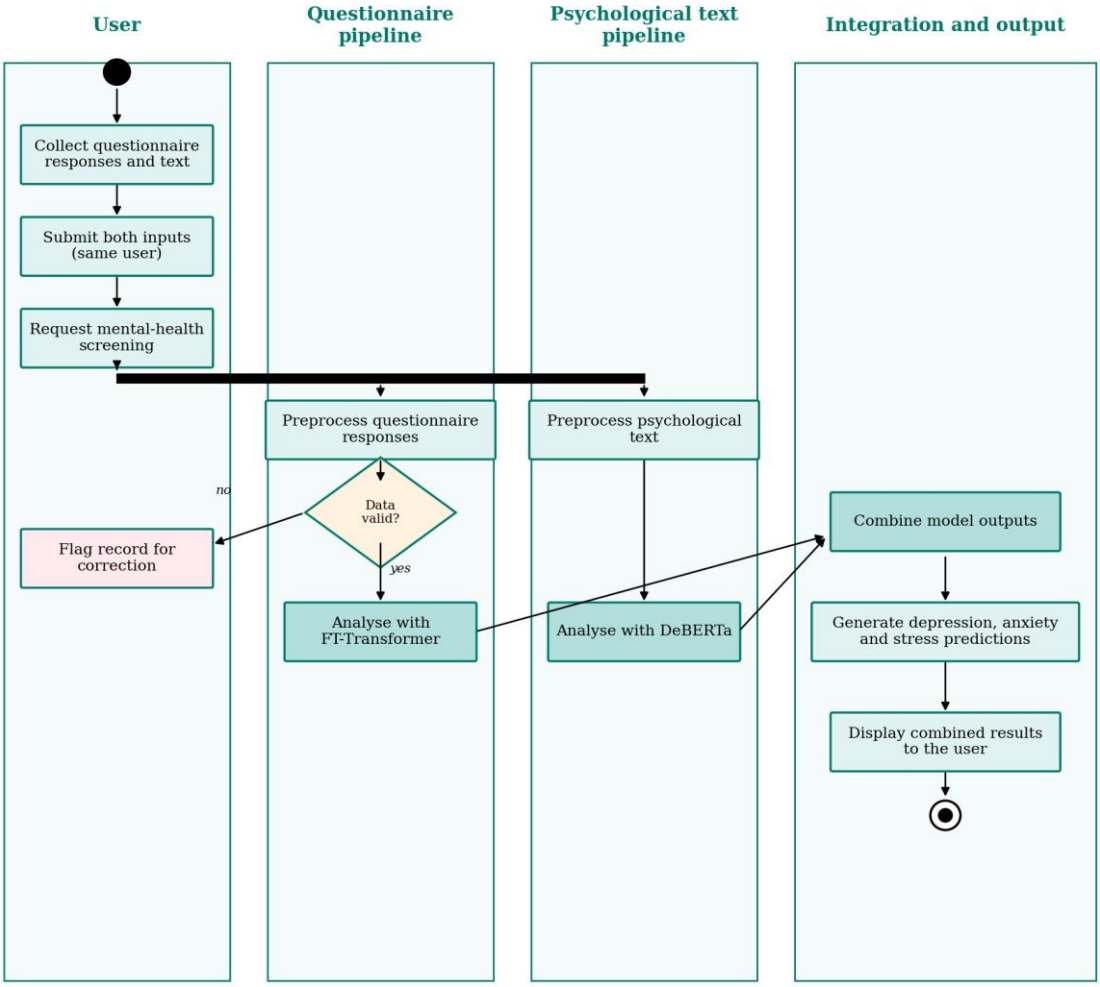


Figure 7.5 : Activity diagram of the mental-health screening process

8. TOOLS/TECHNOLOGIES USED

8.1 Tools

The tools selected for the project are listed in Table 8.1 along with their significance in the work.

Table 8.1 : Tools used and their significance

Tool	Significance in the project
Python	Main programming language for data processing and model development
Pandas	Loading, cleaning and transforming the datasets
NumPy	Numerical operations on arrays and feature matrices
Matplotlib	Plotting of data distributions and results
Seaborn	Statistical visualisation for exploring data quality and relationships
Scikit-learn	Data splitting, evaluation metrics and supporting utilities

8.2 Algorithms

The two analysis models selected for the project are listed in Table 8.2.

Table 8.2 : Algorithms used and their role

Algorithm	Input data	Role in the project
FT-Transformer	Questionnaire data	Analyses structured responses for screening
DeBERTa	Psychological text	Analyses written psychological text for screening

9. CONCLUSION

9.1 Conclusion

This project investigates whether machine-learning models can support early screening for depression, anxiety and stress by combining questionnaire data with psychological text. The FT-Transformer is used for questionnaire analysis and DeBERTa for text analysis, and the outputs of the two models are combined to produce predictions for the three conditions. Predictions are intended as decision-support information for screening and professional review.

The work completed so far includes the following:

- Collection of the questionnaire dataset and the psychological text dataset is complete.
- Model selection is complete: FT-Transformer for questionnaire data and DeBERTa for psychological text.
- The proposed methodology and workflow have been planned.

9.2 Work Plan and Timeline

The work plan is given in Table 9.1. The timeline is provisional and will be updated to match the actual progress and the schedule approved by the guide.

Table 9.1 : Work plan and timeline

Task	Task owner	Status
Problem identification and scope	Team	Done / reviewed
Literature survey and research gaps	Team	Done / reviewed
Project objectives and workplan	Team	Done / reviewed
Requirement analysis and dataset audit	Assigned members	Planned

Task	Task owner	Status
Tools, algorithms and system design	Assigned members	Planned
Proposed methodology and workflow	Team	Planned
Baseline experiments and evaluation	Assigned members	Planned
Phase 1 closure / project exhibition	Team	Planned

9.3 Next Actions

The next actions are to confirm the dataset and permissions, finalise the labels and feature dictionary, implement the baseline models, and present reproducible results at the next review. Data preprocessing is the next step, followed by model development, integration, testing and evaluation.

9.4 Disclosure Statements

- I will maintain academic honesty by doing my own work and avoiding plagiarism, cheating, unauthorised collaboration and the falsification of information.
- I will properly acknowledge and cite all sources I use, including information from the internet, books, software and AI tools.
- I will use institutional resources responsibly and only for authorised academic or research activities. I understand that I am responsible for my own actions and decisions.

10. REFERENCES

[7] Team 5 (GSCSE, GITAM), "Mental Health Analysis Using Machine Learning Models," Capstone Project Introduction, Review 2, GITAM School of CSE, GITAM (Deemed to be University), Visakhapatnam, Oct. 29, 2026.

[1] T. ShamsEldin et al., "Artificial intelligence for predicting depression, anxiety and stress using psychometric data," *Scientific Reports*, vol. 15, Art. no. 37282, 2025, doi: 10.1038/s41598-025-21301-1.

[2] M. D. Nemesure, M. V. Heinz, R. Huang, and N. C. Jacobson, "Predictive modeling of depression and anxiety using electronic health records and a novel machine learning approach with artificial intelligence," *Scientific Reports*, vol. 11, Art. no. 1980, 2021, doi: 10.1038/s41598-021-81368-4.

[3] B. Bailey and D. R. Strunk, "Predicting anxiety and depression over 12 months of the COVID-19 pandemic: A machine learning study," *Journal of Clinical Psychology*, vol. 79, pp. 2388–2403, 2023, doi: 10.1002/jclp.23555.

[4] M. S. Ahmed and N. Ahmed, “A Fast and Minimal System to Identify Depression Using Smartphones: Explainable Machine Learning-Based Approach,” *JMIR Formative Research*, vol. 7, Art. no. e28848, 2023, doi: 10.2196/28848.

[5] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with Disentangled Attention,” arXiv:2006.03654, 2020. [Online]. Available: https://arxiv.org/abs/2006.03654

[6] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, “Revisiting Deep Learning Models for Tabular Data,” arXiv:2106.11959, 2021. [Online]. Available: https://arxiv.org/abs/2106.11959